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Building with AI — Part 2 Interest & Skills Survey

Who wants the advanced session, and what they're ready for

15 student responses · 50-student pool · all 7 teams · July 2026
Mentor responses excluded from all figures

THE HEADLINE

Strong demand, beginner cohort, one clear gap

Demand is real and unanimous

15 of 50 students responded and **every one** is interested — 12 firm "yes," 3 "maybe," zero "no." All 7 teams are represented.

They want the back end

Top requests are **real backends & APIs (73%)**, **databases (67%)**, and **authentication (60%)** — the "missing middle" behind any real product.

The gap = the opportunity

The skills they want most are exactly the ones they're **least confident in**. Backend, cloud & databases sit near-zero on self-rated skill.

Most are building hardware

4 of 7 teams are building physical/IoT products that need a **device → cloud → app** pipeline — the exact stack they're asking to learn.

WHO ANSWERED

15 interested students, all 7 teams

Of a 50-student pool, 15 opted in to the advanced survey. Every respondent wants in — the interest question filtered no one out.

30%

of the 50-student pool responded

15 students · mentors excluded

100%

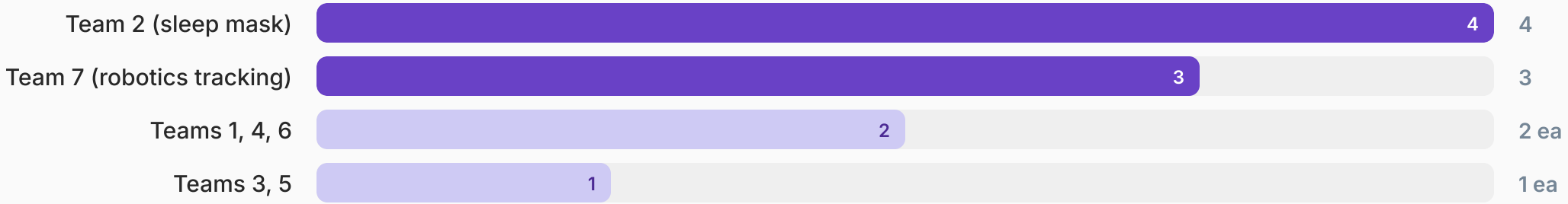
of respondents are interested

12 "yes" · 3 "maybe" · 0 "no"

7 / 7

teams represented

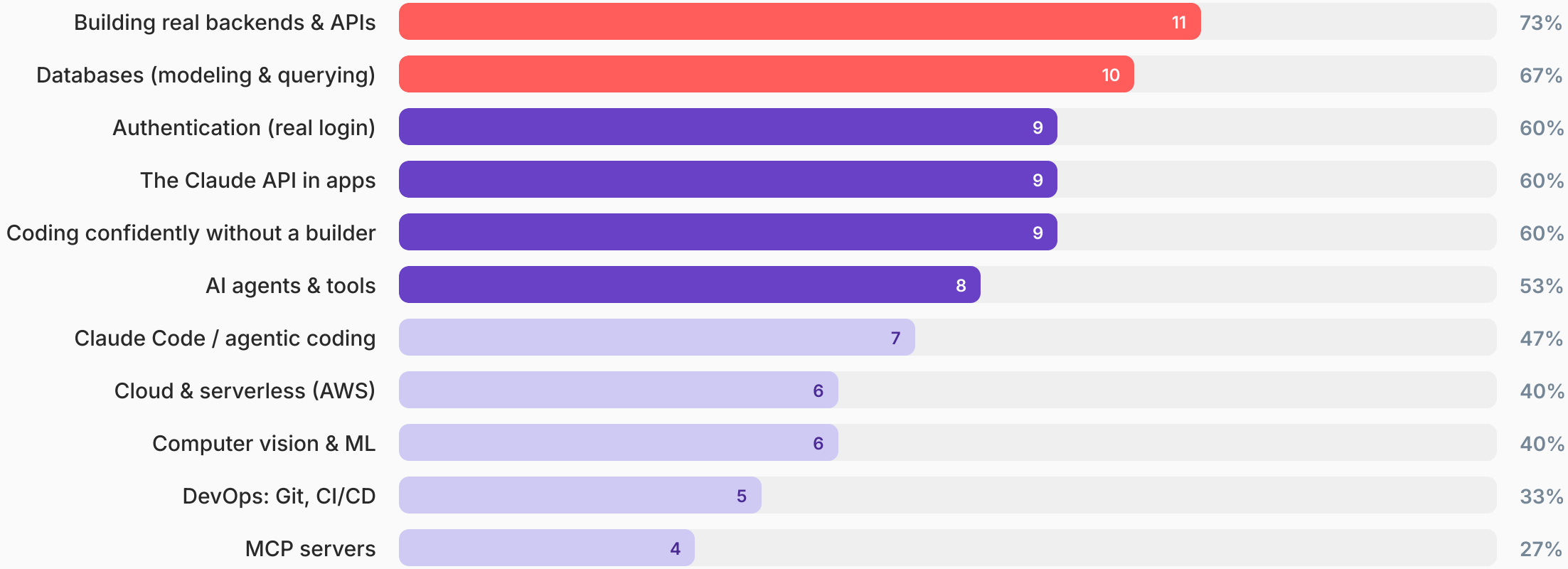
at least one member from every team



DEMAND

What they most want to learn

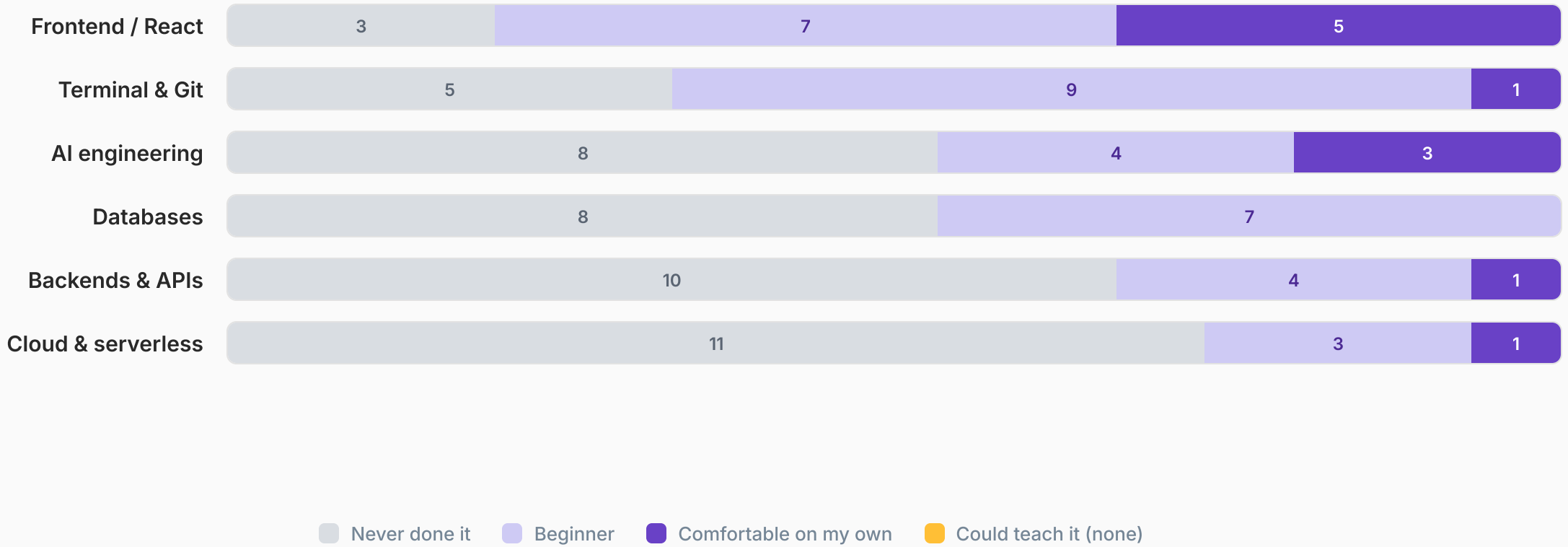
Share of the 15 students selecting each topic. The classic backend stack — APIs, databases, auth — tops the list, followed by the Claude API and coding confidence.



READINESS

Self-rated skill: strong front end, thin everywhere else

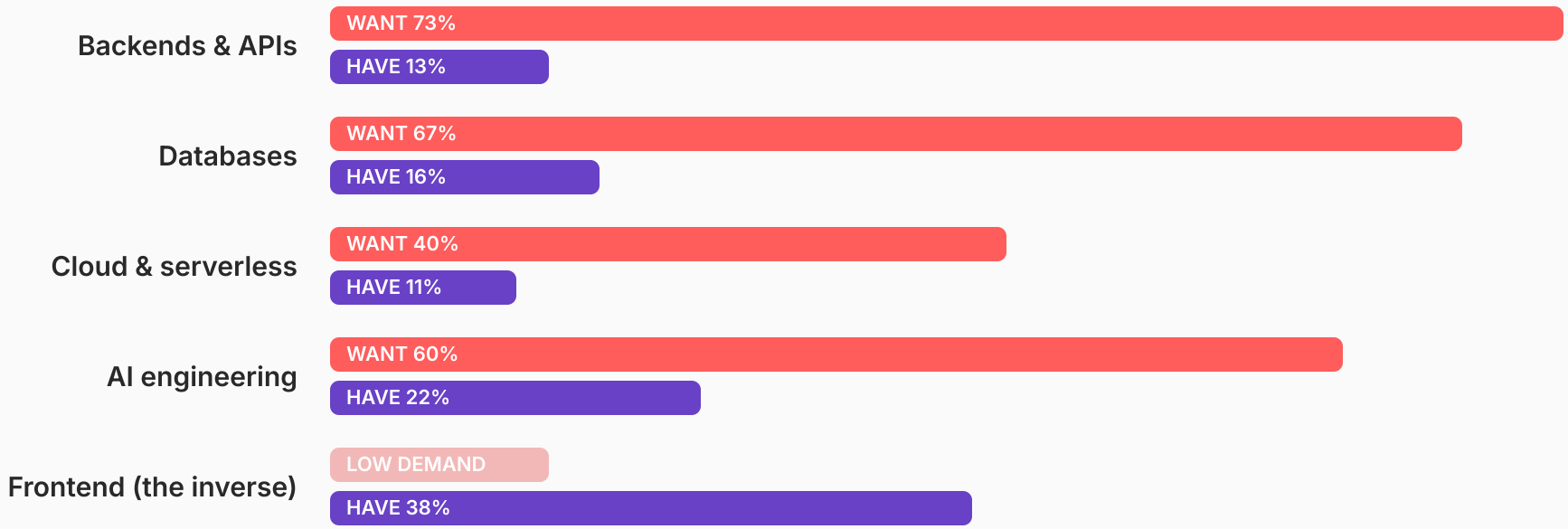
Each bar splits the 15 students by level. Front end is the only area with real comfort; back end, databases and cloud are almost entirely "never done it" or "beginner." Nobody rated themselves "could teach it" in any area.



THE CORE FINDING

They want most what they know least

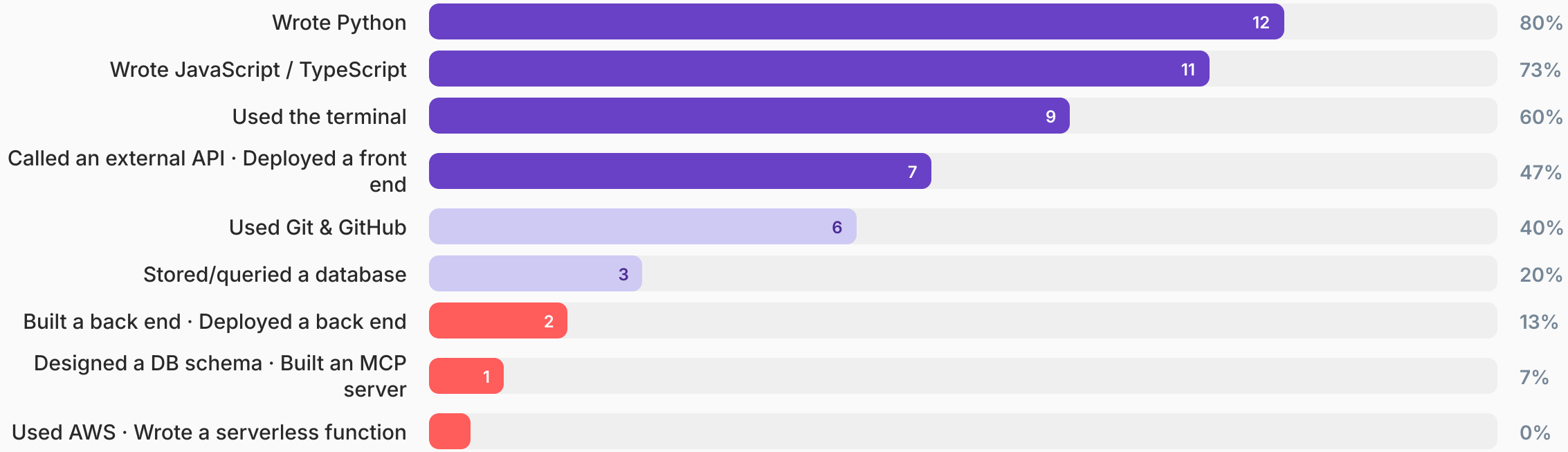
Demand (coral) plotted against current confidence (purple) for each area. The bars point in opposite directions — highest demand meets lowest skill. Front end is the mirror image: real skill, little demand.



REALITY CHECK

Can code and ship a front end — never touched a real back end

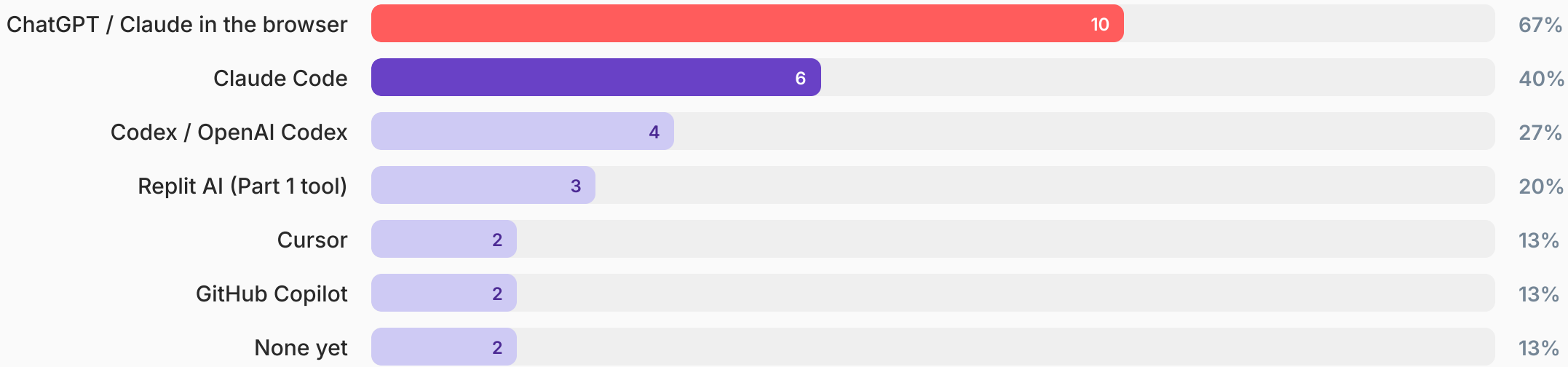
Share of students who've actually done each thing. Language basics and front-end deploys are common; the entire server / cloud / database layer drops off a cliff.



TOOLING

Mostly copy-pasting from a browser

AI coding tools used. Two-thirds live in the ChatGPT/Claude browser copy-paste loop; only 40% have touched an agentic tool like Claude Code — a clear level-up path for the session.



CONFIDENCE

Low-to-mid confidence without an AI builder

"How comfortable are you writing code without an AI builder holding your hand?" (1–5). The cohort clusters low — average 2.5 — and 60% explicitly asked to learn to code more confidently.

2.5

average confidence (of 5)

"needs help" leans over "self-sufficient"

8 / 15

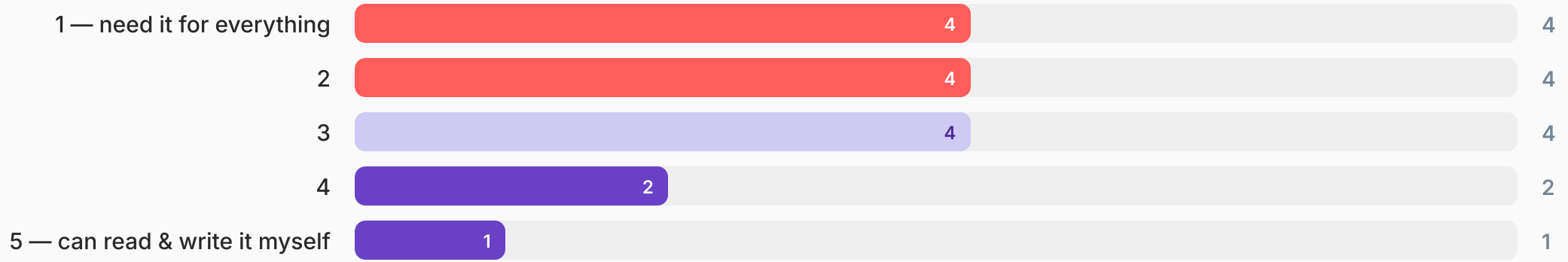
rate themselves 1 or 2

the majority still lean on the tool

1 / 15

rates a full 5

only one fully self-sufficient



THE PRODUCTS

Mostly hardware that needs to talk to an app

4 of the 7 teams are building physical/IoT products. Nearly all need the same backbone: sensor or device → cloud → app. Their stated worries cluster on exactly that hand-off.

1 Dog heat-stroke prevention
Temp sensor + app that calculates safe outdoor time. HARDWARE · IOT

2 Smart sleep mask
Silent vibration alarm, light/noise block, Bluetooth app. HARDWARE · IOT

6 Anti-theft zipper alarm
Alarm on zipper open, iOS app on Find My network. HARDWARE · IOT

7 Robotics workshop tracking
RFID tags + hubs, RSSI to locate lost materials. HARDWARE · IOT

4 AI receptionist + CRM
Answers missed calls for trades, books jobs, captures leads. AI · SOFTWARE

5 Rental verification site
Checks renter-submitted info for apartment owners. SOFTWARE

3 Product e-commerce site
Storefront to sell the team's product. SOFTWARE

! Shared worry: the hand-off
Teams 1, 4, 6 & 7 all flagged connecting device → cloud → app; others flagged hardware assembly.

RECOMMENDATION

Teach the missing middle

Demand, skill gaps and project needs all point the same way: the server / cloud / data layer between a front end and a device. A proposed three-track shape:

1 · Core

Highest demand, lowest skill. Everyone.

- → Real backends & APIs (73% want, 13% have)
- → Databases: modeling & querying (67% / 16%)
- → Authentication done right (60%)
- → Coding without the builder (60%)

2 · Cloud pipeline

Serves the 4 IoT teams directly.

- → Device / app → cloud → data flow
- → AWS basics: Lambda, API Gateway
- → DynamoDB & S3 for storage
- → Deploying a back end that stays up

3 · AI track

60%+ demand; level up the tooling.

- → The Claude API inside an app
- → Agents & tools / function calling
- → Browser copy-paste → Claude Code
- → MCP & CV/ML as electives

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The demand is there. So is the gap.

15 students, every one interested, all asking for the same thing they've never built: real backends, databases, and cloud. An advanced Part 2 that teaches the missing middle meets them exactly where they are.

Next step: confirm session scope & schedule · group by team need · size for ~15–20 seats